

ABNORMALITY REPORT — RCA & CAPA/PAA

Cracked Gas Compressor KO-3201 — High Radial Vibration Trip (Bearing Distress)

AR NUMBER AR-2026-ZCU-0142	EQUIPMENT / TAG KO-3201 (Class A)	PLANT UNIT / DISCIPLINE ZCU / Rotating (ROT)	OCCURRENCE DATE 29 Apr 2026
AR TYPE OPEDR (Operation) — Uptime	SEVERITY LEVEL Catastrophic	PRE-RISK ASSESSMENT II (Score 4000)	RCA LEAD / PIC REL-05

DOWNTIME 32.0 h	PRODUCTION LOSS 1760.0 ton	ESTIMATED LOSS \$1584.0k
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PROBLEM STATEMENT

Cracked Gas Compressor KO-3201 tripped on high-high radial vibration on 29-Apr-2026 due to journal-bearing babbitt distress driven by water-contaminated lube oil.

Immediate Action

Controlled shutdown of compression train, replace DE journal bearing, flush lube-oil system

Chronology of Events

Date / Time	Event	Evidence / Finding
27-Apr-2026	ZCU compression train running normal;	KO-3201 drive-end (DE) radial vibration trending up (from 28 to 52 micron over 5 days).
29-Apr-2026 06:40	KO-3201 DE vibration reached high-high setpoint (75 micron);	machinery protection system tripped the compressor.
29-Apr-2026 08:10	Plant rate cut to minimum;	lube-oil sample taken — water content 1,800 ppm (spec < 500 ppm).
29-Apr-2026 18:00	DE journal bearing removed — babbitt wiped/distressed.	Lube-oil cooler tube leak confirmed.
30-Apr-2026 14:40	New bearing installed, lube-oil system flushed, cooler tube plugged, train restarted to 55 T/H.	

IMPACT SUMMARY (4W)

What: KO-3201 tripped on high-high radial vibration
When: 29-Apr-2026 (06:40)
Scope: ZCU compression train shutdown — full ethylene rate loss
Uptime Loss: 32.0 hours → 1760.0 ton → \$1584.0k
Date Reported: 30 Apr 2026

Past Performance Analysis & Target Setting

HISTORICAL DATA & EVIDENCE

KO-3201 DE vibration baseline is 25–30 micron. Trend shows a steady rise over 5 days preceding the trip. Lube-oil water content had drifted above 500 ppm for ~2 weeks without an online alarm.

TARGET SETTING

ACTUAL CONDITION (Before Improvement)

Compressor trip due to bearing distress from contaminated lube oil

TARGET CONDITION (Project Y)

No compressor trip due to lube-oil contamination (water < 500 ppm, vibration < 45 micron)

Recurrence of this equipment failure mode:
Before = 1 case → Target = 0 case

Parameter Verification (4P) & 4M + 1E Verification

#	Problem Phenomenon / Parameter	Result	Evidence / Finding
P1	Radial vibration above alarm	NG	DE vibration rose to 75 micron vs 45 micron alarm — bearing distress.
P2	Lube-oil water content above spec	NG	Water content 1,800 ppm vs < 500 ppm spec — film breakdown.
P3	Lube-oil supply pressure low	G	Header pressure 1.8 barg — within normal band.
P4	Bearing beyond service life	G	Bearing in service 3 years vs 6-year design — within life.
P5	Process surge / anti-surge margin	G	Anti-surge margin maintained; no surge event recorded.
X1	No online water-in-oil monitoring	NG	Water content only checked by weekly grab sample — slow detection.
X2	Lube-oil cooler integrity not trended	NG	No routine cooler tube leak check between TAs.
X3	Vibration alert limits too wide	NG	Alert at 60 micron gave little lead time before 75 micron trip.
X4	Lube-oil purifier not in service	G	Purifier available and operational.

ROOT CAUSE

Journal-bearing babbitt distress caused by degraded oil-film strength from water ingress via a leaking lube-oil cooler tube, undetected due to absence of online water-in-oil monitoring, leading to rising vibration and a high-high trip.

NG = current condition not meeting standard (problem) • G = meets standard

Impact vs Control Prioritization

CONTROL / IMPACT	Low	Medium	High
Low			
Medium			X1 X2 X3
High			P1 P2

Prioritized root causes (high impact) are addressed first via Corrective & Pro-Active Actions

Corrective & Pro-Active Action (CAPAA)

RC	Action	Corrective / Pro-Active / Preventive Action	Plan Date	PIC	Status
P2	Corrective	Repair leaking lube-oil cooler tube (plug + re-test)	30-Apr-2026	STA-02	Closed
X1	Corrective	Install online water-in-oil sensor on KO-3201 lube system	30-Jun-2026	REL-05	In Progress
P2	Pro-Active	Retube lube-oil cooler at next TA	31-Oct-2026	STA-02	Open
X1	Pro-Active	Apply online water-in-oil monitoring to KO-3202/3203	30-Sep-2026	REL-05	Open
X3	Preventive Vibration limits too wide	Tighten KO-3201 vibration alert to 45 micron & shorten trend-review cadence	30-May-2026	REL-05	
X2	Preventive Cooler integrity not trended	Add lube-oil cooler dP & conductivity trend to reliability dashboard	30-Jun-2026	REL-02	

RISK ANALYSIS OF CORRECTIVE ACTION

Tighten vibration alert to 45 micron

Potential Risk: Nuisance alerts on start-up transients

Countermeasure: Add start-up alarm suppression timer

30-May-2026 | REL-05

PM SCHEDULE ESTABLISHED

PM-1 Weekly	Weekly lube-oil water-content & particle count (Group: Reliability)
PM-2 6-monthly	6-monthly lube-oil cooler leak test (Group: Static)
PM-3 Daily	Continuous vibration trend review KO-3201 (Group: Reliability)

Downtime & Closure Summary

- KEY CORRECTIVE / PRO-ACTIVE ACTIONS
- Repair leaking lube-oil cooler tube (plug + re-test)
 - Install online water-in-oil sensor on KO-3201 lube system
 - Tighten KO-3201 vibration alert to 45 micron & shorten trend-review cadence

DOWNTIME	PRODUCTION LOSS	ESTIMATED LOSS
32.0 h	1760.0 ton	\$1584.0k